

COMP 2121

DISCRETE MATHEMATICS

Quiz 1

Name:

Student ID:

Set: A B C D (circle one)

- Time allowed: 20 min.

1. The System Administrator has set the following rules for the password:

- ✓ The password is a string made of 10 characters.
- ✓ The available characters are M, A, T, H, 1, 2, 3, 4, #
- ✓ Repetition is allowed unless otherwise stated.

a) [1 mark] How many passwords have at most one T?

no T or one T

$$8^{10} + (10) \cdot 8^9$$

any one position for T

b) [1 mark] How many passwords have exactly five M's and exactly three A's?

MMMMMAAA $\xrightarrow{\text{same}}$ M M M M M A A A

$$\frac{10!}{5!3!2!} \cdot 7^2$$

$\left(\frac{10!}{5!3!2!} \right) \cdot 7^2$

c) [1 mark] How many passwords have exactly two T's and two H's, with no T's adjacent, no H's adjacent, and no T next to a H?

select 4 positions

permute

other characters

$$7^6 \cdot \left(\frac{7!}{4!} \right) \cdot \frac{4!}{2!2!}$$

Formulae: $P(n, r) = \frac{n!}{(n-r)!}$

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$$

2. [2 marks] Consider the following programming segment. Your answer must rely on a combination structure.

```

counter = 500
for i = 4 to n+6 {
    counter = counter + 11
}
for j = 4 to 4n {
    for k = j+1 to 3n {
        counter = counter + 15
    }
}
// assume n>5

```

Handwritten notes:

- For the first loop: $n+6-4+1 = n+3$
- For the second loop: $* \text{no gap!}$, $3n-4+1 = 3n-3$

Determine the value of the variable *counter* after the code segment executes. Provide your answer as a formula in terms of *n*.

Note: The for loops include both the start and end values.

$$\text{counter} = 500 + (n+3)(11) + \binom{3n-3}{2} \cdot 15$$

Formulae: $P(n, r) = \frac{n!}{(n-r)!}$

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

$$\binom{n+r-1}{r} = \frac{(n+r-1)!}{r!(n-1)!}$$